

Organizational Mental Models: How Firms Think and Form Beliefs

Executive Summary

Organizations think. Not in the way individuals think, but through distributed processes of **sensemaking** — forming collective beliefs, updating strategic assumptions, and constructing mental models of the business environment. Under Active Inference, cognition is the process of **posterior inference**: updating beliefs about hidden states given new observations. This module examines how organizations build and maintain their generative models, why those models sometimes resist updating, and how cognitive pathologies (groupthink, cognitive rigidity, sunk cost thinking) emerge.

Learning Objectives

1. Explain organizational cognition as **belief updating** under the generative model
 2. Identify the **strategic assumptions** (priors) that drive your organization's decisions
 3. Understand **sensemaking** (Weick) as the organizational process of posterior inference
 4. Diagnose cognitive pathologies: groupthink, belief rigidity, and the sunk cost trap
 5. Apply **assumption auditing** to reveal hidden beliefs that shape strategy
-

Key Concepts

1. The Organizational Generative Model

Every organization operates on a set of beliefs about how the world works:

Belief Category	Example	If Wrong, Consequences
Customer beliefs	"Our customers value reliability over price"	Losing price-sensitive segments to competitors
Market beliefs	"This market will grow 10% annually"	Over-investment in a flat or declining market
Competitive beliefs	"Our main competitor is Company X"	Blindsided by disruption from unexpected entrant
Capability beliefs	"We are an innovation leader"	Resting on past laurels while capabilities decay
Causal beliefs	"Advertising drives sales"	Misallocating marketing budget

These beliefs constitute the organization's **generative model** — the systematic framework from which all predictions flow.

2. Sensemaking: Organizational Posterior Inference

Karl Weick's concept of **sensemaking** maps directly onto Active Inference:

- **Observation:** Ambiguous or unexpected signal arrives (e.g., quarterly sales drop)
- **Noticing:** Someone in the organization flags the signal as meaningful
- **Interpretation:** Teams construct candidate explanations ("market downturn," "competitor launched similar product," "our product quality declined")
- **Belief updating:** The organization selects the explanation that best accounts for the evidence and adjusts its generative model accordingly
- **Action:** Strategy is adjusted based on the updated model

Case Study — Nokia: Nokia's sensemaking apparatus failed catastrophically in 2007–2010. When the iPhone launched, Nokia's internal model held that "phones are hardware businesses where scale and cost matter most." Early signals that smartphones were becoming software platforms were interpreted through this existing model — leading to the conclusion that the

iPhone was a niche luxury product, not a category-defining shift. The organization's generative model resisted updating because the implied strategic change was too threatening.

3. Priors: Strategic Assumptions That Resist Change

Priors are the beliefs the organization holds *before* new evidence arrives. Strong priors are valuable when they reflect deep expertise — but dangerous when they become rigid:

- **Healthy priors:** "In our industry, R&D investment pays off over 5-year cycles" (based on decades of evidence)
- **Pathological priors:** "Electric vehicles will never achieve mainstream adoption" (Kodak-level denial of change)

The strength of a prior (its precision) determines how much evidence is needed to change it. Organizations with **very strong priors** require overwhelming evidence before updating — which often means they update too late.

4. Cognitive Pathologies

Pathology	Mechanism	Organizational Example
Groupthink	High precision on consensus, low precision on dissent	Bay of Pigs invasion — advisors suppressed doubts
Belief rigidity	Priors so strong that no evidence can change them	Blockbuster's model that physical rental would persist
Sunk cost trap	Continuing to invest in a failing strategy because of past investment	Concorde project — continued despite uneconomic fundamentals
Confirmation bias	Selectively attending to confirming evidence	Theranos — internal culture that punished skepticism
Anchoring	Over-weighting initial information in all subsequent estimates	Budget processes anchored to last year's numbers

5. Assumption Auditing

A structured process for making implicit beliefs explicit and evaluating their validity:

1. **Elicit:** Ask "What do we believe to be true about our market/customers/competitors?"
 2. **Categorize:** Classify assumptions as testable vs. untestable, critical vs. non-critical
 3. **Challenge:** For each critical assumption, ask "What evidence would change this belief?"
 4. **Design tests:** Create epistemic actions (experiments, research) to test key assumptions
 5. **Schedule reviews:** Revisit assumptions quarterly, not just during annual planning
-

Applications

Cognitive Audit: Your Organization's Mental Model

Map the key beliefs driving your organization's strategy:

Belief	Strength (1–5)	Last Updated	Evidence Base	Vulnerability

Cross-References

- For shared cognition in teams, see [Collective Intelligence: Cognition](#)
- For strategic assumption analysis, see [Strategic Modeling: Cognition](#)
- For ML inference and hybrid cognition, see [Digital Transformation: Cognition](#)

Summary

Concept	Organizational Meaning
Generative model	The set of beliefs driving organizational decisions
Sensemaking	The collective process of interpreting ambiguous signals
Priors	Strategic assumptions held before new evidence arrives
Posterior inference	Updating beliefs based on new observations
Cognitive pathologies	Groupthink, rigidity, sunk costs, confirmation bias

References

- Weick, K. E. (1995). *Sensemaking in Organizations*. Sage.

- Argyris, C. (1977). Double loop learning in organizations. *Harvard Business Review*, 55(5), 115–125.
- Parr, T., Pezzulo, G., & Friston, K. J. (2022). *Active Inference*. MIT Press.